# Mohsen Zare

### Senior Full-Stack Engineer · TypeScript, Node.js, React · LLM product features

Leiden, Netherlands (from February 2027) · Currently Yerevan, Armenia

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Work permit: Highly Skilled Migrant or EU Blue Card through the employer, switched inside the Netherlands (about 2 weeks with a recognised sponsor). No relocation needed.

## Summary

Senior full-stack engineer with 7+ years building production web and mobile products, the last four in TypeScript: NestJS/Node.js APIs on PostgreSQL, React/Next.js and React Native (Expo) front ends, AWS. I build LLM features that ship: retrieval with citations, AI assistants grounded on product data, and AI search with rule-based fallback. I lead a six-person engineering team day to day and act as product manager, and I have built and run three production platforms on my own, from database schema to deployment.

## Skills

- **Languages:** TypeScript, JavaScript, Python, SQL
- **Backend:** Node.js, NestJS, Express, REST, OpenAPI/Swagger, Prisma, Zod, JWT/OTP auth, rate limiting, scheduled jobs
- **Frontend & mobile:** React, Next.js, React Native, Expo, Vue 3, Nuxt 3, PWA, web performance, design systems
- **AI / LLM:** LLM-powered features, RAG with citations, pgvector, PostgreSQL full-text search, AI assistants, prompt pipelines, Google Gemini, OpenAI API, speech-to-text (Whisper)
- **Data:** PostgreSQL (schema design, indexing, migrations), Supabase, DynamoDB, MySQL
- **Cloud & DevOps:** AWS (Lambda, API Gateway, S3, CloudFront, WAF, CDK), Cloudflare, Docker, GitHub Actions, CI/CD, Linux, Nginx
- **Testing:** Jest, Vitest, Supertest, Playwright, end-to-end tests
- **Ways of working:** API design, code review, technical specs, product management, Agile/Scrum, Shape Up
- **Also:** PHP/Laravel (earlier roles)

## Professional Experience

**Senior Full Stack Developer — Bayzian (bernoly.com)**

Oct 2024 – Present · Tallinn, Estonia (remote)

B2B SaaS. Day-to-day lead of a six-person engineering team across frontend, backend and infrastructure.

- Lead delivery for the team: planning, code review, design sign-off and QA; also act as product manager for prioritisation, specs and release planning.
- Migrated the core data layer from DynamoDB to PostgreSQL, moving the product to a relational model.
- Built LLM-powered product features end to end, from Node.js services to the React/TypeScript UI; redesigned the in-product AI assistant panel of the agent studio.
- Added AWS CloudFront as edge gateway/CDN and restructured the frontend build (code splitting, caching), cutting page load times by about 40%.
- Fixed workflow-timer and human-in-the-loop bugs, and closed a security gap that let clients forge system messages.
- Introduced a docs-before-code workflow (Shape Up) with git and CI hooks in the frontend and backend repositories, so every feature starts from a reviewed spec.

**CTO & Co-founder — TailStory (part-time)**

2025 – Present · Amsterdam (remote) · tailstory-app.com

AI pet-health record: a photo becomes a structured pet profile, and health questions are answered with citations from veterinary sources.

- Built and run the whole stack on my own: NestJS API on PostgreSQL/Supabase; one Expo (React Native) codebase for iOS, Android and the web (web app live); app and Next.js site in 21 languages.
- Built the AI pipeline: Gemini extraction from photo to profile, and retrieval over 21 veterinary textbooks with pgvector and PostgreSQL full-text search. Answers cite their sources; keyword search answers in 0.2–1.2 s in production.
- Added phone (SMS) and Google sign-in, push notifications, scheduled jobs, rate limiting, an OpenAPI contract and a Jest/Supertest suite; I deploy with GitHub Actions to a Linux VPS behind Cloudflare.

**Full Stack Developer — Mirorim**
Apr 2024 – Oct 2024 · Tehran, Iran

- Built backend services and REST APIs with NestJS and PostgreSQL, including the relational schemas for core features.
- Delivered the matching Vue/Nuxt front end, owning features from API contract to release.
- In parallel, a contract for SDStudio (Oct 2024 – Jan 2025): Vue/Nuxt interfaces on existing services.

**Full Stack Developer / Frontend Lead — IREEN**
Oct 2021 – Mar 2024 · Vienna, Austria (remote)

- Promoted to Frontend Lead within eight months; owned the frontend architecture of a real-estate valuation platform and worked on its API and data layer.
- Designed a reusable component system used across the product.
- Led performance and integration work in an agile/Scrum team.

**Full Stack Developer — SISHEMI / University of Tehran Science & Technology Park**
Feb 2020 – Sep 2021 · Tehran, Iran

- Built and maintained production web applications end to end, from API endpoints to UI; promoted to senior.

**PHP / Laravel & Vue Developer**
2017 – 2020 · Tehran, Iran

- Laravel, Vue and MySQL development for several companies, freelance WordPress work, and teaching assistance in PHP.

## Selected Projects

**DibaMart — e-commerce platform for security equipment (own venture, in production)**

NestJS, Prisma, PostgreSQL, Next.js, Expo, Whisper

- One API contract serves a Next.js storefront and admin panel (PWA) and an Expo app.
- AI shopping assistant grounded on the catalogue, courses, brands, warranty terms and an admin-editable knowledge base, with optional cited web search.
- Product search with instant and AI modes (AI falls back to rules on failure) and voice search through self-hosted speech-to-text.
- Four-tier supplier pricing with per-supplier credit ledgers, and order tracking with SMS escalation to suppliers on a 4-hour SLA.

**HamHam — pet-food brand platform (client project, in production)**

Next.js 16, NestJS, Prisma, Zod

- Monorepo with shared Zod contracts; website, admin panel and API, with SMS one-time-password login and an AI nutrition chat.
- 330+ automated tests (90 unit and 41 end-to-end API tests, 202 web tests); Lighthouse 100 for accessibility, best practices and SEO; 272 legacy URLs redirected.

## Education

Associate Degree (Kardani) in Agriculture — Shahid Chamran University of Ahvaz, Iran

Equivalent to a Dutch Associate degree (NLQF level 5). Software engineering is self-taught and proven through 7+ years of professional work.

## Languages

English (professional working proficiency) · Persian (native)